

INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

DIGITAL FORENSIC INVESTIGATION REPORT

"Preparing Global Cybersecurity Leaders Through Excellence in Hands-On Training"

Case Title: Case Study 1: Investigating Harassment Email Traffic With Wireshark	Course Name: SBT-DF204 Computer Forensics Case Studies
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Operating Platform: Kali Linux 2026 (Rolling Release)	Date of Submission: 5/10/2026
Case Scope / Scenario: NITROBA University Harassment Incident	Report Version: Version 1.0 (Final Release)

1. Executive Summary

This report documents the forensic investigation of a harassment email submitted through the anonymous web service www.willselfdestruct.com. The investigation acquired and preserved a 54MB packet capture (nitroba.pcap), identified the client device (192.168.15.4, MAC 00:17:f2:e2:c0:ce), and reconstructed the harassment POST request. Through analysis of concurrent HTTP session cookies, the device was linked to the Gmail account jcoachj@gmail.com, which corresponds to Johnny Coach, a student on the Chemistry 109 roster. The evidence strongly supports attribution, though limitations of the shared open wireless network are documented.

2. Authority, Scope, and Evidence Description

- **Authority:** This investigation was conducted using the authorized NITROBA University Harassment Scenario training capture.
- **Forensic Scope:** Analysis of HTTP traffic to identify the source of a harassment email sent to lilytuckrige@yahoo.com.
- **Primary Evidence Artifact:** nitroba.pcap (SHA-256: 2b77a9eafc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb).
- **Working Forensic Duplicate:** nitroba_working.pcap (Verified bit-identical hash matching the original capture).

3. Environment and Acquisition Method

- **Host Operating System:** Kali Linux 2026 (Rolling Release).
- **Packet Dissection Suite:** TShark / Wireshark 4.6.6 with specialized display filters and TCP stream reassembly.
- **Capture Verification:** SHA-256 cryptographic verification before and after all forensic filtering operations.

4. Timeline of Events

Chronological event log tracking suspect activity across the NITROBA University wireless network (EDT, UTC-04:00):

Time (EDT)	Forensic Event Description	Supporting Packet / Evidence
02:00:44	Gmail session active (jcoachj@gmail.com) on device 192.168.15.4	Cookie: OL_SESSION=jcoachj@gmail.com-cal
02:01:26	Device visits sendanonymousemail.net (prior attempt)	HTTP GET / Frame 80614
02:02:57	POST to sendanonymousemail.net/send.php	Frame 80614 (Failed/aborted submission)
02:03:43	Device navigates to willselfdestruct.com	GET /secure/submit
02:04:24	Harassment POST transmitted to willselfdestruct.com/secure/submit	Frame 83601 (TCP Stream 1707)
02:04:24	Server responds with 302 Moved Temporarily	Location: http://www.willselfdestruct.com/secure/success
02:04:55	Gmail session activity continues on same device	Cookie remains active across Google channels

5. Analysis and Findings

Evidentiary Payload: 5.1 Harassment Message Content

To: lilytuckrige@yahoo.com
Subject: you can't find us
Message: and you can't hide from us.
Stop teaching.
Start running.

Reconstructed from Frame 83601 (TCP Stream 1707) via HTTP form field parsing (to, subject, message, ttl=30, submit.x=92, submit.y=26).

Hardware Signature: 5.2 Device Attribution

- **Client IP Address:** 192.168.15.4 (Ephemeral Source Port: 36044).
- **Destination Server:** 69.25.94.22 (HTTP Port 80, www.willselfdestruct.com).
- **Client Hardware MAC Address:** 00:17:f2:e2:c0:ce (Consistent across 34,554 captured frames).
- **Client Software User-Agent:** Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1).

Account Correlation: 5.3 Identity Attribution

The device at 192.168.15.4 was concurrently authenticated to Google services using cookies OL_SESSION=jcoachj@gmail.com-cal and gmailchat=jcoachj@gmail.com/475090. This active session spanned from 02:00:44 through 02:04:55 EDT, encompassing the precise moment of the harassment POST at 02:04:24 EDT.

Institutional Match: 5.4 Roster Correlation

Cross-referencing the email identifier with the Chemistry 109 course roster (evidence/roster.txt) identified Johnny Coach as an enrolled student. The username jcoachj directly maps to J. Coach (Johnny Coach). The unrelated

Facebook cookie (beth@bethr.org) did not match any entry on the course roster and represents another user on the open campus network.

6. Answers to Investigation Questions

Definitive answers to the core forensic investigation prompts:

Investigative Question	Forensic Finding / Answer
1. File acquired & preserved?	nitroba.pcap (SHA-256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb); working copy verified identical.
2. Client system?	192.168.15.4:36044 communicating with server 69.25.94.22:80.
3. Evidence linking to message?	HTTP POST body in Frame 83601 with exact recipient, subject, and threat text.
4. Hardware device?	Ethernet MAC address 00:17:f2:e2:c0:ce.
5. Associated account / user?	Active Google session jcoachj@gmail.com mapped to Johnny Coach.
6. Present on Chem 109 roster?	YES — Johnny Coach is confirmed on the Chemistry 109 student roster.
7. Timestamp of incident?	2008-07-22 02:04:24 EDT.
8. Defensible forensic conclusion?	HIGH CONFIDENCE: Johnny Coach submitted the harassment email from device 192.168.15.4.

7. Challenges and Limitations

- **Shared Open Wireless Network:** Multiple users share the wireless local area network; however, concurrent persistent session cookies directly link the Gmail account to the specific MAC address at the exact second of transmission.
- **MAC Address / User-Agent Discrepancy:** The MAC OUI prefix (00:17:f2) belongs to Apple Inc., whereas the User-Agent identifies Windows XP / IE6. While this can indicate MAC spoofing, it is also consistent with running Windows within a virtual machine or Boot Camp on Apple hardware.
- **Absence of Physical Endpoint Forensics:** Investigation was strictly limited to passive network packet captures; host disk or memory forensics could not be performed.
- **Red Herring Identification:** The Facebook cookie beth@bethr.org observed on the network belongs to an unrelated party and is discarded as non-probative.

8. Recommendations

- **Enforce TLS / HTTPS Exclusively:** Deprecate plaintext HTTP transmission across campus networks to ensure web applications and mail submission services encrypt session data.
- **Implement Network Access Control (802.1X):** Deploy WPA2/WPA3-Enterprise with 802.1X RADIUS authentication to tie every hardware MAC address directly to authenticated user credentials.
- **Filter Anonymous Mail Portals:** Configure perimeter firewall and web proxy rules to block or log access to anonymous message services like willselfdestruct.com and sendanonymousemail.net.
- **Correlate Multi-Source Telemetry:** In operational investigations, pair network packet captures with wireless controller association logs and DHCP lease histories for ironclad attribution.

9. References

- Digital Corpora. (2008). NITROBA University Harassment Scenario Packet Capture. Retrieved from <https://digitalcorpora.org/>
- Wireshark Foundation. (2023). Wireshark User Guide. Retrieved from <https://www.wireshark.org/docs/>
- National Institute of Standards and Technology (NIST). (2021). Guide to Digital Forensics (Special Publication 800-86). U.S. Department of Commerce.
- International Cybersecurity and Digital Forensics Academy (ICDFA). (2026). Computer Forensics Case Studies Courseware.

10. Appendix – Commands and Evidence Log

Evidence Matrix: Appendix A: Formal Evidence Inventory Log:

ID	Finding Description	Evidence Source	Investigative Significance
E01	Harassment POST Content	Frame 83601	Proves message text & target
E02	TCP Stream 1707 Conversation	Stream 1707	Request & response headers
E03	Client Hardware MAC	00:17:f2:e2:c0:ce	Device hardware identification
E04	Gmail Session Cookie	jcoachj@gmail.com	Identity attribution
E05	Roster Search Match	Johnny Coach	Institutional enrollment correlation
E06	Capture SHA-256 Hash	2b77a9eafc1...	Chain of custody integrity
E07	Facebook Cookie (Red Herring)	beth@bethr.org	Unrelated campus network user
E08	Prior Anonymous Mail Attempt	sendanonymousemail.net	Demonstrates premeditation/intent

Command Strings: Appendix B: Key Commands Executed:

```
# Capture Integrity Hashing
sha256sum evidence/nitroba.pcap working/nitroba_working.pcap

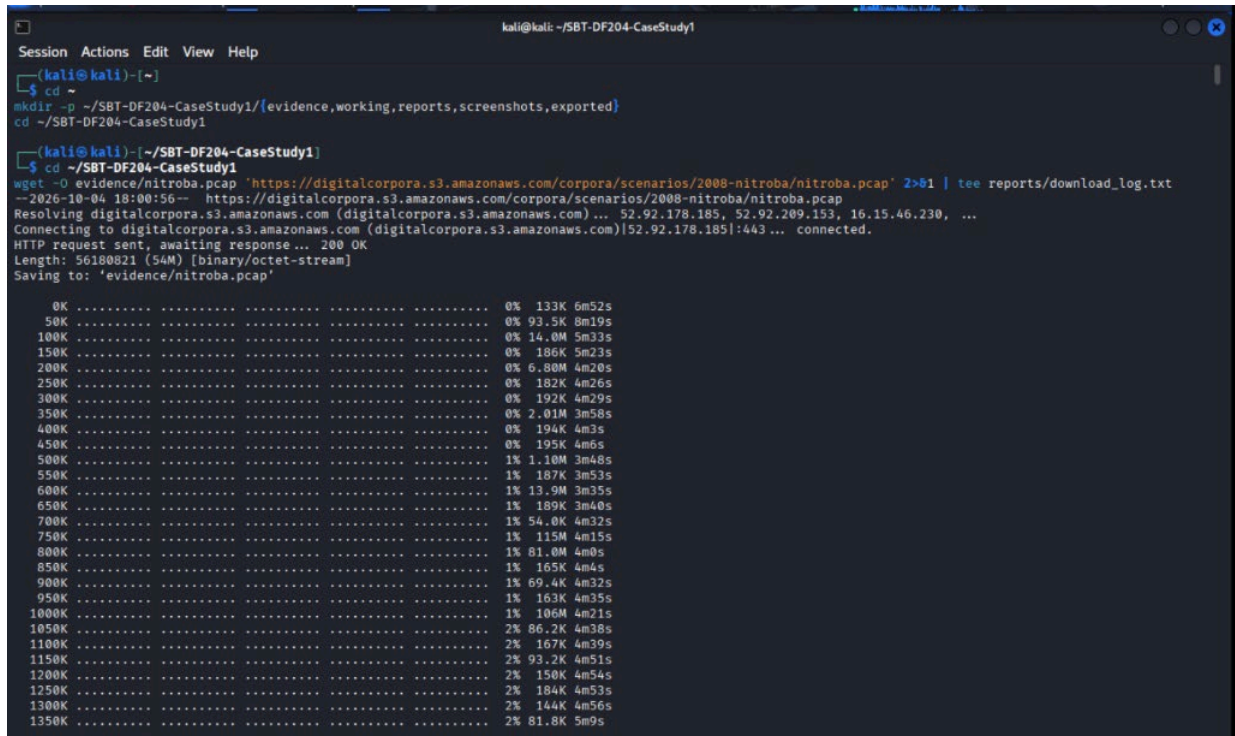
# Identify Harassment Service & Reconstruct Stream 1707
tshark -r working/nitroba_working.pcap -Y 'http.request.method == "POST"' -T fields ...
tshark -r working/nitroba_working.pcap -q -z follow,tcp,ascii,1707

# Extract Candidate MAC Address & Cookies
tshark -r working/nitroba_working.pcap -Y 'ip.src == 192.168.15.4 && eth.src' -T fields -e eth.src |
sort | uniq -c
tshark -r working/nitroba_working.pcap -Y 'ip.src == 192.168.15.4 && http.cookie' -T fields -e
http.cookie | grep -i '@'

# Course Roster Search
grep -i -E 'jcoach|coach' evidence/roster.txt
```

11. Appendix C: Screenshots and Evidence

The following 18 documented figures depict the complete investigative progression from evidence download and protocol hierarchy analysis through stream reconstruction, identity attribution, and final case conclusions:



```
kali@kali: ~/SBT-DF204-CaseStudy1
Session Actions Edit View Help
(kali@kali)~$ cd ~
$ mkdir -p ~/SBT-DF204-CaseStudy1/{evidence,working,reports,screenshots,exported}
cd ~/SBT-DF204-CaseStudy1

(kali@kali)~/SBT-DF204-CaseStudy1$ cd ~/SBT-DF204-CaseStudy1
$ wget -O evidence/nitroba.pcap 'https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap' 2>&1 | tee reports/download_log.txt
--2026-10-04 18:00:56-- https://digitalcorpora.s3.amazonaws.com/corpora/scenarios/2008-nitroba/nitroba.pcap
Resolving digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)... 52.92.178.185, 52.92.209.153, 16.15.46.230, ...
Connecting to digitalcorpora.s3.amazonaws.com (digitalcorpora.s3.amazonaws.com)|52.92.178.185|:443 ... connected.
HTTP request sent, awaiting response... 200 OK
Length: 56180821 (54M) [binary/octet-stream]
Saving to: 'evidence/nitroba.pcap'

 0K ..... 0% 133K 6m52s
 50K ..... 0% 93.5K 8m19s
100K ..... 0% 14.0M 5m33s
150K ..... 0% 186K 5m23s
200K ..... 0% 6.80M 4m20s
250K ..... 0% 182K 4m26s
300K ..... 0% 192K 4m29s
350K ..... 0% 2.01M 3m58s
400K ..... 0% 194K 4m3s
450K ..... 0% 195K 4m6s
500K ..... 1% 1.10M 3m48s
550K ..... 1% 187K 3m53s
600K ..... 1% 13.9M 3m35s
650K ..... 1% 189K 3m40s
700K ..... 1% 54.0K 4m32s
750K ..... 1% 115M 4m15s
800K ..... 1% 81.0M 4m0s
850K ..... 1% 165K 4m4s
900K ..... 1% 69.4K 4m32s
950K ..... 1% 163K 4m35s
1000K ..... 1% 106M 4m21s
1050K ..... 2% 86.2K 4m38s
1100K ..... 2% 167K 4m39s
1150K ..... 2% 93.2K 4m51s
1200K ..... 2% 150K 4m54s
1250K ..... 2% 184K 4m53s
1300K ..... 2% 144K 4m56s
1350K ..... 2% 81.8K 5m9s
```

Figure 1: Evidence Download & Workspace Setup — Downloading nitroba.pcap (54MB) from Digital Corpora repository into /SBT-DF204-CaseStudy1 directory.

```
kali@kali: ~/SBT-DF204-CaseStudy1
Session Actions Edit View Help

(kali@kali)~/SBT-DF204-CaseStudy1
$ ls -lh evidence/nitroba.pcap
file evidence/nitroba.pcap
-rw-rw-r-- 1 kali kali 54M Nov 21 2020 evidence/nitroba.pcap
evidence/nitroba.pcap: pcap capture file, microsecond ts (little-endian) - version 2.4 (Ethernet, capture length 4096)

(kali@kali)~/SBT-DF204-CaseStudy1
$ cd ~/SBT-DF204-CaseStudy1
sha256sum evidence/nitroba.pcap | tee reports/nitroba_sha256.txt
cat reports/nitroba_sha256.txt
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb evidence/nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb evidence/nitroba.pcap

(kali@kali)~/SBT-DF204-CaseStudy1
$ cp --preserve=timestamps evidence/nitroba.pcap working/nitroba_working.pcap
sha256sum evidence/nitroba.pcap working/nitroba_working.pcap | tee reports/working_copy_hashes.txt
cat reports/working_copy_hashes.txt
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb evidence/nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb working/nitroba_working.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb evidence/nitroba.pcap
2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb working/nitroba_working.pcap

(kali@kali)~/SBT-DF204-CaseStudy1
$ capinfos working/nitroba_working.pcap | tee reports/capinfos.txt
cat reports/capinfos.txt
File name: working/nitroba_working.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 4096 bytes
Number of packets: 94 k
File size: 56 MB
Data size: 54 MB
Capture duration: 15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time: 2008-07-22 02:13:47.046029
Data byte rate: 3,468 bytes/s
Data bit rate: 27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
```

inside or press Ctrl+G

Figure 2: Capture File Integrity Hashing & Metadata Verification — Preserving nitroba.pcap, computing SHA-256 hash (2b77a9ea...), duplicating to working copy, and running capinfos (94k packets).

```
kali@kali: ~/SBT-DF204-CaseStudy1
Session Actions Edit View Help
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 4096
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 94410
File name: working/nitroba_working.pcap
File type: Wireshark/tcpdump/... - pcap
File encapsulation: Ethernet
File timestamp precision: microseconds (6)
Packet size limit: file hdr: 4096 bytes
Number of packets: 94 k
File size: 56 MB
Data size: 54 MB
Capture duration: 15759.950751 seconds
Earliest packet time: 2008-07-21 21:51:07.095278
Latest packet time: 2008-07-22 02:13:47.046029
Data byte rate: 3,468 bytes/s
Data bit rate: 27 kbps
Average packet size: 579.07 bytes
Average packet rate: 5 packets/s
SHA256: 2b77a9eaefc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
SHA1: 65656392412add15f93f8585197a8998aaeb50a1
Strict time order: True
Number of interfaces in file: 1
Interface #0 info:
    Encapsulation = Ethernet (1 - ether)
    Capture length = 4096
    Time precision = microseconds (6)
    Time ticks per second = 1000000
    Number of stat entries = 0
    Number of packets = 94410

(kali@kali)-[~/SBT-DF204-CaseStudy1]
$ tshark -r working/nitroba_working.pcap -q -z io,phs | tee reports/protocol_hierarchy.txt
cat reports/protocol_hierarchy.txt
Warning: program compiled against libxml 215 using older 214
```

Figure 3: *Capinfos Capture Statistics & Duration Dissection* — Detailed capture properties: 94,410 packets, 15,759.95s duration, 27 kbps data rate, and Ethernet encapsulation.

```
(kali㉿kali)-[~/SBT-DF204-CaseStudy1]
$ tshark -r working/nitroba_working.pcap -q -z io,phs | tee reports/protocol_hierarchy.txt
cat reports/protocol_hierarchy.txt
Warning: program compiled against libxml 215 using older 214
```

Protocol Hierarchy Statistics
Filter:

frame	frames:94410 bytes:54670237
eth	frames:94410 bytes:54670237
ip	frames:90379 bytes:54413507
tcp	frames:83114 bytes:52515171
http	frames:10103 bytes:7007064
image-gif	frames:1136 bytes:737880
image-jfif	frames:1180 bytes:982489
data-text-lines	frames:744 bytes:529526
xml	frames:581 bytes:554874
png	frames:136 bytes:112564
http	frames:4 bytes:1484
urlencoded-form	frames:28 bytes:14515
media	frames:467 bytes:248381
data	frames:7 bytes:7097
amf	frames:1 bytes:511
ocsp	frames:2 bytes:1108
tls	frames:1561 bytes:931191
tls	frames:216 bytes:245641
vssmonitoring	frames:1 bytes:64
_ws.malformed	frames:1 bytes:64
prp	frames:2 bytes:134
xmpp	frames:18 bytes:3131
data	frames:50 bytes:12304
msnms	frames:6 bytes:820
ymsg	frames:8 bytes:2219
prp	frames:1 bytes:191
ymsg	frames:1 bytes:375
ymsg	frames:1 bytes:375
ymsg	frames:1 bytes:375
udp	frames:7200 bytes:1874348
dns	frames:2905 bytes:434973
sip	frames:297 bytes:201582
ntp	frames:184 bytes:17296

Figure 4: Overall Protocol Hierarchy Statistics (Top Level) — Dissecting capture protocols: Ethernet, IPv4 (90,379 frames), TCP (83,114 frames), HTTP (10,103 frames), and UDP.

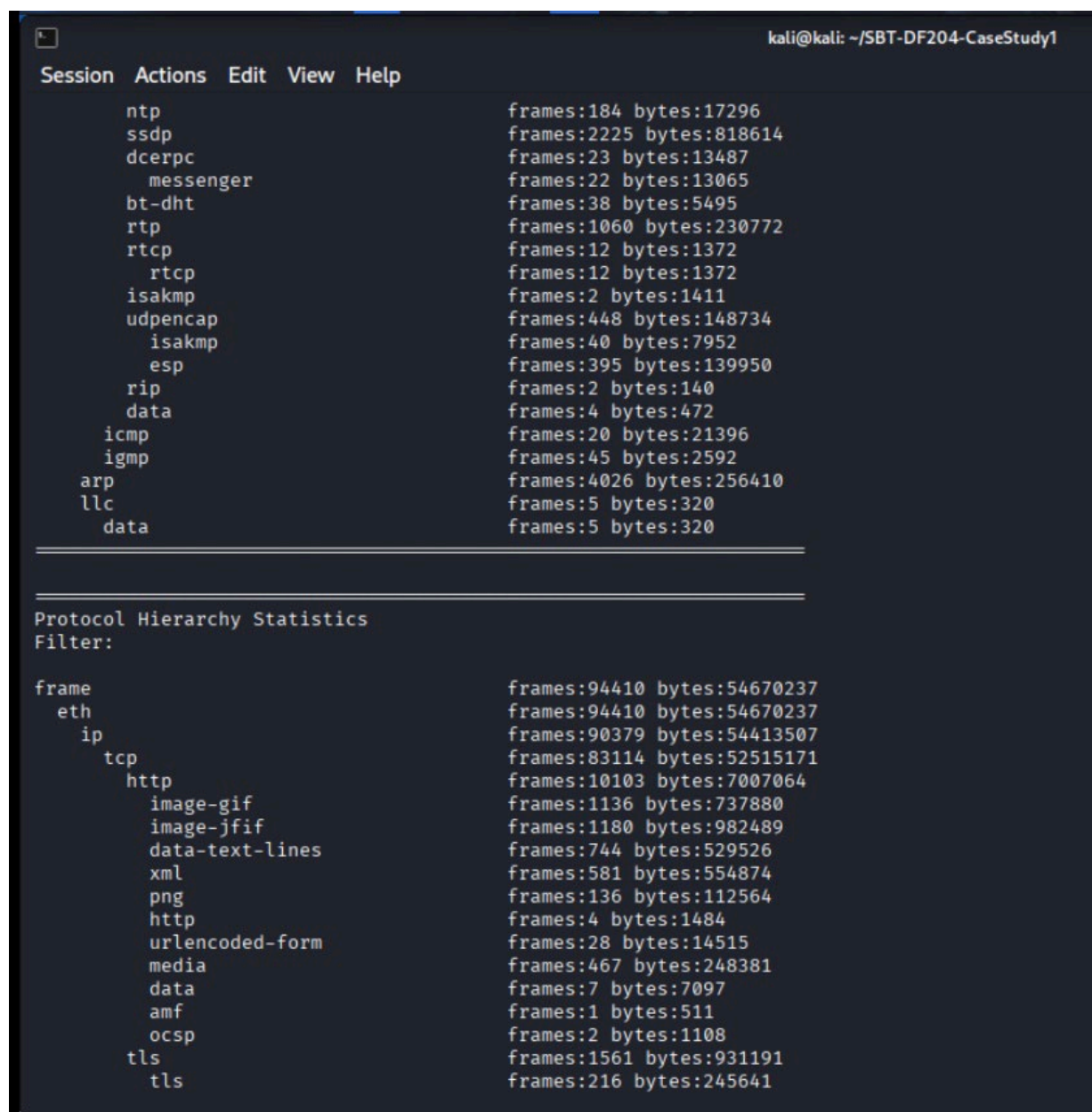


Figure 5: *Protocol Hierarchy Statistics (Application Services)* — Tracing higher layers: HTTP sub-elements, TLS, SSDP, DCERPC, RTP, ICMP, and ARP traffic.

```

(kali@kali) [~/SBT-DF204-CaseStudy1]
$ cd ~/SBT-DF204-CaseStudy1
CAP-working/nitroba_working pcap
tshark -r "$CAP" -Y 'http.request.method == "GET" && http.host contains "willself" -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e http.host
t -e http.request.uri 2>/dev/null | tee reports/willself_get.txt
cat reports/willself_get.txt
82936 2008-07-22T02:03:43.825871000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /secure/submit
82985 2008-07-22T02:03:44.125949000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/spacer.gif
83025 2008-07-22T02:03:44.334254000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/sm-logo.gif
83037 2008-07-22T02:03:44.374583000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/warning-home.gif
83072 2008-07-22T02:03:44.530352000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/body-bk.gif
83087 2008-07-22T02:03:44.599079000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/bttm-send.gif
83162 2008-07-22T02:03:44.852265000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/bridge_small.gif
83614 2008-07-22T02:04:24.564165000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /secure/success
83654 2008-07-22T02:04:24.902157000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/bk-message.gif
82936 2008-07-22T02:03:43.825871000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /secure/submit
82985 2008-07-22T02:03:44.125949000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/spacer.gif
83025 2008-07-22T02:03:44.334254000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/sm-logo.gif
83037 2008-07-22T02:03:44.374583000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/warning-home.gif
83072 2008-07-22T02:03:44.530352000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/body-bk.gif
83087 2008-07-22T02:03:44.599079000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/bttm-send.gif
83162 2008-07-22T02:03:44.852265000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/bridge_small.gif
83614 2008-07-22T02:04:24.564165000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /secure/success
83654 2008-07-22T02:04:24.902157000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /images/bk-message.gif

(kali@kali) [~/SBT-DF204-CaseStudy1]
$ tshark -r "$CAP" -Y 'http.request.method == "POST" -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e http.host -e http.request.uri 2>/dev
/null | tee reports/http_posts.txt
cat reports/http_posts.txt
10217 2008-07-21T21:56:23.170447000-0400 192.168.1.64 64.12.187.25 sync.aol.com /m57jean
10241 2008-07-21T21:56:23.335872000-0400 192.168.1.64 64.12.187.25 sync.aol.com /m57jean
10248 2008-07-21T21:56:23.580097000-0400 192.168.1.64 64.12.187.25 sync.aol.com /m57jean
10869 2008-07-21T21:59:54.595252000-0400 192.168.1.64 64.4.21.61 www.update.microsoft.com /v6/UpdateRegulationService/UpdateRegulatio
n.aspx
17299 2008-07-21T23:44:46.722832000-0400 192.168.1.64 74.125.19.19 mail.google.com /mail/?ui=26ik=4233eca8e46view=au8rt=sj
17305 2008-07-21T23:44:48.890808000-0400 192.168.1.64 74.125.19.19 mail.google.com /mail/?ui=26ik=4233eca8e46view=cv8th=11b48086a8a331ad5prf=1
6init=16nsc=16rt=j6search=inbox
17316 2008-07-21T23:44:49.893979000-0400 192.168.1.64 74.125.19.19 mail.google.com /mail/?ui=26ik=4233eca8e46view=cv8th=11b482d8bc502422d6prf=1
6nsc=16rt=j6search=inbox
17373 2008-07-21T23:44:55.683957000-0400 192.168.1.64 74.125.19.19 mail.google.com /mail/channel/bind?at=xn3j3022s94pi22yqiqnx80qxagfx78VER=6b
it=23596RID=53996CVER=66zxxkg7qakm9396t=1
17384 2008-07-21T23:44:57.766721000-0400 192.168.1.64 74.125.19.19 mail.google.com /mail/channel/bind?at=xn3j3022s94pi22yqiqnx80qxagfx78VER=6b

```

Figure 6: Identifying Web Traffic to willselfdestruct.com — Filtering HTTP GET requests from IP 192.168.15.4 accessing /secure/submit on 69.25.94.22.

```

Session Actions Edit View Help
73354 2008-07-22T01:57:50.747302000-0400 192.168.15.4 75.126.76.142 vodpod.com /blogplayer.swf?v=32326file=http://web1.vodpod.com:80/video
/trans/3/103/3103/flv/1203.flv&autoStart=false&thumb=http://s3.amazonaws.com/vodpod.com/videos/thumbnull
74344 2008-07-22T01:58:07.909706000-0400 192.168.15.4 65.55.13.158 www.update.microsoft.com /v6/UpdateRegulationService/UpdateRegulatio
n.aspx
77710 2008-07-22T02:00:45.360749000-0400 192.168.15.4 74.125.19.104 www.google.com /calendar/calddetails
77711 2008-07-22T02:00:45.361069000-0400 192.168.15.4 74.125.19.104 www.google.com /calendar/load
79097 2008-07-22T02:01:05.186648000-0400 192.168.15.4 74.125.19.17 mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf6ui=16V
ER=46RID=422486CVER=36zx=2bem6fi8b3oh6it=454
79141 2008-07-22T02:01:05.937762000-0400 192.168.15.4 74.125.19.17 mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf6ui=16V
ER=26SID=67F8DC1634D9D3136RID=422496zx=fgtn2bw9qazv6it=782
79187 2008-07-22T02:01:06.373943000-0400 192.168.15.4 74.125.19.17 mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf6ui=16V
ER=26SID=67F8DC1634D9D3136RID=422505zx=pgks2dsuufm6it=203
79652 2008-07-22T02:01:13.690666000-0400 192.168.15.4 74.125.19.17 mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf6ui=16V
ER=26SID=67F8DC1634D9D3136RID=422515zx=6nu1yeej5n8g6it=250
79732 2008-07-22T02:01:17.225783000-0400 192.168.15.4 74.125.19.17 mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf6ui=16V
ER=26SID=67F8DC1634D9D3136RID=422526zx=61f17kk9h7776it=3719
79749 2008-07-22T02:01:17.425424000-0400 192.168.15.4 74.125.19.104 www.google.com /calendar/calddetails
79750 2008-07-22T02:01:17.425760000-0400 192.168.15.4 74.125.19.104 www.google.com /calendar/load
80025 2008-07-22T02:02:05.562559000-0400 192.168.15.4 74.125.19.17 mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf6ui=16V
ER=26SID=67F8DC1634D9D3136RID=422536zx=teggawway3et26it=52125
80614 2008-07-22T02:02:57.548149000-0400 192.168.15.4 69.80.225.91 www.sendanonymousemail.net /send.php
80842 2008-07-22T02:03:05.557263000-0400 192.168.15.4 74.125.19.17 mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf6ui=16V
ER=26SID=67F8DC1634D9D3136RID=422546zx=9tduxqkx4td66it=112125
83326 2008-07-22T02:04:05.546193000-0400 192.168.15.4 74.125.19.17 mail.google.com /mail/channel/bind?at=xn3j32oktf2a0q6oa3k9sfr6d09yzf6ui=16V
ER=26SID=67F8DC1634D9D3136RID=422555zx=ws8oir6zf6iy6it=172125
83601 2008-07-22T02:04:24.311700000-0400 192.168.15.4 69.25.94.22 www.willselfdestruct.com /secure/submit
90504 2008-07-22T02:09:59.850757000-0400 192.168.15.4 209.191.110.214 filetransfer.msg.yahoo.com:80 /notifyft
94263 2008-07-22T02:11:35.320531000-0400 192.168.15.4 199.7.71.72 ocsip.thawte.com:80 /

(kali@kali) [~/SBT-DF204-CaseStudy1]
$ tshark -r "$CAP" -Y 'data-text-lines contains "secure anonymous E-mail" || data-text-lines contains "lily" || data-text-lines contains "Lily" -T field
s -e frame.number -e ip.src -e ip.dst -e http.host -e data-text-lines 2>/dev/null | head -30 | tee reports/harassment_content_search.txt
cat reports/harassment_content_search.txt
82998 69.25.94.22 192.168.15.4 Line-based text data: text/html (539 lines)
83641 69.25.94.22 192.168.15.4 Line-based text data: text/html (429 lines)
82998 69.25.94.22 192.168.15.4 Line-based text data: text/html (539 lines)
83641 69.25.94.22 192.168.15.4 Line-based text data: text/html (429 lines)

(kali@kali) [~/SBT-DF204-CaseStudy1]
$

```

Figure 7: Inventorying HTTP POST Traffic Across Network — Identifying client IP 192.168.15.4 issuing POST to sendanonymousemail.net (Frame 80614) and willselfdestruct.com (Frame 83601).

```
kali@kali: ~/SBT-DF204-CaseStudy1
Session Actions Edit View Help
(kali@kali)~(~/SBT-DF204-CaseStudy1)
$ cd ~/SBT-DF204-CaseStudy1
CAP=working/nitroba_working.pcap

# Get the TCP stream number for frame 83601
tshark -r "$CAP" -Y 'frame.number == 83601' -T fields -e tcp.stream 2>/dev/null | tee reports/harassment_stream_number.txt
cat reports/harassment_stream_number.txt
1707
1707

(kali@kali)~(~/SBT-DF204-CaseStudy1)
$ STREAM=$(cat reports/harassment_stream_number.txt | tr -d '[:space:]')
echo "Following stream: 1707"
tshark -r "$CAP" -q -z follow,tcp,ascii,1707 2>/dev/null | tee reports/harassment_stream.txt
cat reports/harassment_stream.txt
Following stream: 1707

=====
Follow: tcp,ascii
Filter: tcp.stream eq 1707
Node 0: 192.168.15.4:36044
Node 1: 69.25.94.22:80
649
POST /secure/submit HTTP/1.1
Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */*
Referer: http://www.willselfdestruct.com/secure/submit
Accept-Language: en-us
Content-Type: application/x-www-form-urlencoded
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)
Host: www.willselfdestruct.com
Content-Length: 188
Connection: Keep-Alive
Cache-Control: no-cache

to=lilytuckrige@yahoo.com&from=subject=you+can%27t+find+us&message=and+you+can%27t+hide+from+us.%0D%0A%0D%0Astop+teaching.%0D%0A%0D%0Astart+running.+&type
=0&ttl=30&submit.x=92&submit.y=26
289
HTTP/1.1 302 Moved Temporarily
Date: Tue, 22 Jul 2008 07:24:45 GMT
Server: Apache
Set-Cookie: JSESSIONID=C16BFAC8710DA82AB3D8D80AE0F4F05A; Path=/
Location: http://www.willselfdestruct.com/secure/success
```

Figure 8: TCP Stream 1707 Reassembly & Harassment POST Extraction — Reconstructing full HTTP POST transaction in Frame 83601 revealing harassment text to lilytuckrige@yahoo.com.

```
kali@kali: ~/SBT-DF204-CaseStudy1
Session Actions Edit View Help
Location: http://www.willselfdestruct.com/secure/success
Content-Length: 0
Connection: close
Content-Type: text/plain; charset=UTF-8

=====
Follow: tcp,ascii
Filter: tcp.stream eq 1707
Node 0: 192.168.15.4:36044
Node 1: 69.25.94.22:80
649
POST /secure/submit HTTP/1.1
Accept: image/gif, image/x-bitmap, image/jpeg, image/pjpeg, application/x-shockwave-flash, */*
Referer: http://www.willselfdestruct.com/secure/submit
Accept-Language: en-us
Content-Type: application/x-www-form-urlencoded
Accept-Encoding: gzip, deflate
User-Agent: Mozilla/4.0 (compatible; MSIE 6.0; Windows NT 5.1; SV1)
Host: www.willselfdestruct.com
Content-Length: 188
Connection: Keep-Alive
Cache-Control: no-cache

to=lilytuckrige@yahoo.com&from=subject=you+can%27t+find+us&message=and+you+can%27t+hide+from+us.%0D%0A%0D%0Astop+teaching.%0D%0A%0D%0Astart+running.+&type
=0&ttl=30&submit.x=92&submit.y=26
289
HTTP/1.1 302 Moved Temporarily
Date: Tue, 22 Jul 2008 07:24:45 GMT
Server: Apache
Set-Cookie: JSESSIONID=C16BFAC8710DA82AB3D8D80AE0F4F05A; Path=/
Location: http://www.willselfdestruct.com/secure/success
Content-Length: 0
Connection: close
Content-Type: text/plain; charset=UTF-8

(kali@kali)~(~/SBT-DF204-CaseStudy1)
$ tshark -r "$CAP" -Y 'frame.number == 83601' -T fields -e http.file_data 2>/dev/null | tee reports/post_body.txt
```

Figure 9: Server HTTP 302 Redirection & Post Body File Export — Server responding 302 Moved Temporarily to /secure/success and extracting raw payload bytes to post_body.txt.


```

(kali@kali)~[~/SBT-DF204-CaseStudy1]
$ tshark -r "$CAP" -Y "frame.number == 83601 || (http.cookie contains 'jcoachj' && ip.src == 192.168.15.4)" -T fields -e frame.number -e frame.time -e ip.src -e ip.dst -e http.host 2>/dev/null | tee reports/correlation_summary.txt
cat reports/correlation_summary.txt | head -10
77528 2008-07-22T02:01:44.478524000-0400 192.168.15.4 74.125.19.104 www.google.com
77547 2008-07-22T02:01:44.668574000-0400 192.168.15.4 74.125.19.104 www.google.com
77710 2008-07-22T02:01:45.269749000-0400 192.168.15.4 74.125.19.104 www.google.com
77711 2008-07-22T02:01:45.361069000-0400 192.168.15.4 74.125.19.104 www.google.com
77727 2008-07-22T02:01:50.381580000-0400 192.168.15.4 74.125.19.104 www.google.com
78967 2008-07-22T02:01:02.113436000-0400 192.168.15.4 74.125.19.17 mail.google.com
78968 2008-07-22T02:01:02.115094000-0400 192.168.15.4 74.125.19.17 mail.google.com
78975 2008-07-22T02:01:02.337389000-0400 192.168.15.4 74.125.19.17 mail.google.com
78977 2008-07-22T02:01:02.377906000-0400 192.168.15.4 74.125.19.17 mail.google.com
78984 2008-07-22T02:01:02.466796000-0400 192.168.15.4 209.85.201.189 chatenabled.mail.google.com
78990 2008-07-22T02:01:02.547046000-0400 192.168.15.4 74.125.19.17 mail.google.com
78996 2008-07-22T02:01:02.588700000-0400 192.168.15.4 74.125.19.17 mail.google.com
79007 2008-07-22T02:01:02.761943000-0400 192.168.15.4 74.125.19.17 mail.google.com
79012 2008-07-22T02:01:02.800863000-0400 192.168.15.4 74.125.19.17 mail.google.com
79014 2008-07-22T02:01:02.823811000-0400 192.168.15.4 209.85.201.189 b.mail.google.com
79019 2008-07-22T02:01:02.975899000-0400 192.168.15.4 74.125.19.17 mail.google.com
79023 2008-07-22T02:01:03.023427000-0400 192.168.15.4 74.125.19.17 mail.google.com
79032 2008-07-22T02:01:03.079754000-0400 192.168.15.4 74.125.19.17 mail.google.com
79036 2008-07-22T02:01:03.184192000-0400 192.168.15.4 74.125.19.17 mail.google.com
79040 2008-07-22T02:01:03.295597000-0400 192.168.15.4 74.125.19.17 mail.google.com
79044 2008-07-22T02:01:03.389552000-0400 192.168.15.4 74.125.19.17 mail.google.com
79049 2008-07-22T02:01:03.504461000-0400 192.168.15.4 74.125.19.17 mail.google.com
79053 2008-07-22T02:01:03.596041000-0400 192.168.15.4 74.125.19.17 mail.google.com
79057 2008-07-22T02:01:03.711645000-0400 192.168.15.4 74.125.19.17 mail.google.com
79061 2008-07-22T02:01:03.798909000-0400 192.168.15.4 74.125.19.17 mail.google.com
79065 2008-07-22T02:01:03.918448000-0400 192.168.15.4 74.125.19.17 mail.google.com
79069 2008-07-22T02:01:04.006313000-0400 192.168.15.4 74.125.19.17 mail.google.com
79073 2008-07-22T02:01:04.132430000-0400 192.168.15.4 74.125.19.17 mail.google.com
79077 2008-07-22T02:01:04.214321000-0400 192.168.15.4 74.125.19.17 mail.google.com
79087 2008-07-22T02:01:04.626483000-0400 192.168.15.4 74.125.19.17 mail.google.com
79097 2008-07-22T02:01:05.186648000-0400 192.168.15.4 74.125.19.17 mail.google.com
79101 2008-07-22T02:01:05.438953000-0400 192.168.15.4 74.125.19.17 mail.google.com
79113 2008-07-22T02:01:05.472647000-0400 192.168.15.4 74.125.19.17 mail.google.com
79117 2008-07-22T02:01:05.514963000-0400 192.168.15.4 209.85.201.189 b.mail.google.com
79132 2008-07-22T02:01:05.695419000-0400 192.168.15.4 74.125.19.17 mail.google.com
79141 2008-07-22T02:01:05.937762000-0400 192.168.15.4 74.125.19.17 mail.google.com
79142 2008-07-22T02:01:05.938415000-0400 192.168.15.4 74.125.19.17 mail.google.com
79154 2008-07-22T02:01:06.137714000-0400 192.168.15.4 74.125.19.17 mail.google.com

```

Figure 12: Persistent Gmail Session Evidence Trace — Listing consecutive Google session frames confirming continuous jcoachj authentication across the incident window.

```

(kali@kali)~[~/SBT-DF204-CaseStudy1]
$ # URL-decode the POST body
python3 << EOF | tee reports/harassment_message_decoded.txt
import urllib.parse
body = "to:lilytuckrige@yahoo.com&from=&subject=you+can%27+find+us&message=and+you+can%27+hide+from+us.%0D%0A%0D%0AStop+teaching.%0D%0A%0D%0AStart+runnin
g.&type=0&ttl=30&submit.x=92&submit.y=26"
parsed = urllib.parse.parse_qs(body)
print("== Decoded Harassment POST Body ==")
for k, v in parsed.items():
    print(f"{k} = {v[0]}")
EOF
cat reports/harassment_message_decoded.txt
== Decoded Harassment POST Body ==
to = lilytuckrige@yahoo.com
subject = you can't find us
message = and you can't hide from us.

Stop teaching.

Start running.
type = 0
ttl = 30
submit.x = 92
submit.y = 26
== Decoded Harassment POST Body ==
to = lilytuckrige@yahoo.com
subject = you can't find us
message = and you can't hide from us.

Stop teaching.

Start running.
type = 0
ttl = 30
submit.x = 92
submit.y = 26

(kali@kali)~[~/SBT-DF204-CaseStudy1]
$ tshark -r "$CAP" -Y 'ip.src == 192.168.15.4 && (http.host contains "willself" || http.host contains "anonymousmail" || http.host contains "gmail" || h
ttp.host contains "facebook")' -T fields -e frame.number -e frame.time -e ip.dst -e http.host -e http.request.uri 2>/dev/null | tee reports/suspicious_time
line.txt
cat reports/suspicious_timeline.txt | head -30
31951 2008-07-22T00:50:28.119394000-0400 69.63.178.11 facebook.com /

```

Figure 13: Python URL-Decoding of Harassment Payload — Executing Python script to decode URL-encoded form parameters: to, subject, and message lines.

```

(kali@kali)-[~/SBT-DF204-CaseStudy1]
$ tshark -r "$CAP" -Y 'ip.src == 192.168.15.4 && (http.host contains "willself" || http.host contains "anonymousemail" || http.host contains "gmail" || http.host contains "facebook")' -T fields -e frame.number -e frame.time -e ip.dst -e http.host -e http.request.uri 2>/dev/null | tee reports/suspicious_time_line.txt
cat reports/suspicious_time_line.txt | head -30
31951 2008-07-22T00:50:28.119394000-0400 69.63.178.11 facebook.com /
31959 2008-07-22T00:50:28.237438000-0400 69.63.176.40 www.facebook.com /
32202 2008-07-22T00:50:30.257741000-0400 69.63.176.40 www.facebook.com /images/upload_progress.gif
32229 2008-07-22T00:50:30.689396000-0400 69.63.176.40 www.facebook.com /ajax/precog.php?email=beth%40bethr.org60.9555108418343606
32284 2008-07-22T00:50:35.834540000-0400 69.63.176.40 www.facebook.com /home.php?
32492 2008-07-22T00:50:38.181638000-0400 128.241.218.25 photos-g.ak.facebook.com /photos-ak-sctm/v43/214/2425101550/app_2_2425101550_2298.gi
f
32515 2008-07-22T00:50:38.228720000-0400 128.241.218.24 photos-b.ak.facebook.com /photos-ak-sctm/v43/65/15015611585/app_2_15015611585_7183.g
if
32517 2008-07-22T00:50:38.229924000-0400 128.241.218.24 photos-b.ak.facebook.com /photos-ak-sctm/v43/65/18199467785/app_2_18199467785_5172.g
if
32528 2008-07-22T00:50:38.243661000-0400 128.241.218.17 photos-a.ak.facebook.com /photos-ak-sctm/v43/40/2357179312/app_2_2357179312_2638.gif
32529 2008-07-22T00:50:38.243944000-0400 128.241.218.17 photos-a.ak.facebook.com /photos-ak-sctm/v43/212/2601240224/app_2_2601240224_8511.gi
f
32540 2008-07-22T00:50:38.266868000-0400 208.111.148.6 photos-538.ll.facebook.com /photos-ll-sctm/v43/178/2558160538/app_2_2558160538_3048.gi
f
32550 2008-07-22T00:50:38.286866000-0400 128.241.218.81 photos-e.ak.facebook.com /photos-ak-sctm/v43/188/6921127772/app_2_6921127772_3737.gi
f
32552 2008-07-22T00:50:38.288875000-0400 128.241.218.81 photos-e.ak.facebook.com /photos-ak-sctm/v43/116/4188419540/app_2_4188419540_9544.gi
f
32560 2008-07-22T00:50:38.312382000-0400 68.142.79.70 photos-544.ll.facebook.com /photos-ll-sctm/v43/240/26047640544/app_2_26047640544_4369.
gif
32564 2008-07-22T00:50:38.319808000-0400 128.241.218.17 photos-a.ak.facebook.com /photos-ak-sctm/v43/0/2405948328/app_2_2405948328_5147.gif
32575 2008-07-22T00:50:38.348757000-0400 69.22.167.247 ads.ak.facebook.com /ads2/flyers/65/29/6002209355230_1_f6eb04ee.jpg
32576 2008-07-22T00:50:38.349844000-0400 204.2.133.42 profile.ak.facebook.com /object2/1105/104/t8993367613_1183.jpg
32577 2008-07-22T00:50:38.350416000-0400 204.2.133.42 profile.ak.facebook.com /v222/1282/26/q1271531631_74.jpg
32598 2008-07-22T00:50:38.431369000-0400 68.142.123.254 photos-915.ll.facebook.com /photos-ll-sctm/v43/191/7629233915/app_2_7629233915_7653.gi
f
32621 2008-07-22T00:50:38.520029000-0400 204.2.133.42 profile.ak.facebook.com /v52/1040/112/q589120240_9284.jpg
32652 2008-07-22T00:50:38.608568000-0400 204.2.133.42 profile.ak.facebook.com /v222/309/33/q19791_2987.jpg
32714 2008-07-22T00:50:38.758065000-0400 69.63.176.40 www.facebook.com /swf/SoundPlayer.swf
32724 2008-07-22T00:50:38.855685000-0400 69.63.176.174 0.channel14.facebook.com:80 /iframe/97rstatic.ak.fbcdn.net/rsrc.php/pkg/60/109886/js/c
ommon.js.pkg.php
32740 2008-07-22T00:50:40.827922000-0400 69.63.176.174 0.channel14.facebook.com /x/2327026504/true/p_588141158=08
32763 2008-07-22T00:51:00.952739000-0400 69.63.176.40 www.facebook.com /reqs.php?quickling=true
32852 2008-07-22T00:51:03.119688000-0400 204.2.133.42 profile.ak.facebook.com /object2/88/85/s22105098592_9793.jpg
32855 2008-07-22T00:51:03.128870000-0400 204.2.133.42 profile.ak.facebook.com /v226/335/50/s1294760555_8949.jpg

```

Figure 14: Facebook Cookie Inspection (beth@bethr.org - Red Herring) — Identifying unrelated Facebook session cookies on 192.168.15.4 belonging to another campus user.

```

(kali@kali)-[~/SBT-DF204-CaseStudy1]
$ cd ~/SBT-DF204-CaseStudy1

# Search for jcoach, coach, jcoachj, etc.
grep -i -E "jcoach|coach|jcoachj|coachj" evidence/roster.txt | tee reports/roster_matches_jcoach.txt
cat reports/roster_matches_jcoach.txt

# Search for beth, bethr, bethr.org
grep -i -E "beth|bethr" evidence/roster.txt | tee reports/roster_matches_beth.txt
cat reports/roster_matches_beth.txt

# If the above return nothing, try searching for the domain or any email-like strings
grep -i "@ " evidence/roster.txt | tee reports/roster_emails.txt
cat reports/roster_emails.txt | head -30
• Johnny Coach
• Johnny Coach

(kali@kali)-[~/SBT-DF204-CaseStudy1]
$ head -20 evidence/roster.txt
Chemistry 109 - Class roster
Teacher: Lily Tuckrige
Students:
• Amy Smith
• Burt Greedom
• Tuck Gorge
• Ava Book
• Johnny Coach
• Jeremy Ledvkin
• Nancy Colburne
• Tamara Perkins
• Esther Pringle
• Asar Misrad
• Jenny Kant

```

Figure 15: Chemistry 109 Roster Search (Johnny Coach Correlation) — Searching roster.txt via grep confirming Johnny Coach enrolled under Teacher Lily Tuckrige.

```

└─$ cd ~/SBT-DF204-CaseStudy1
cat > reports/final_timeline.txt << 'EOF'
=== NITROBA HARASSMENT CASE - FORENSIC TIMELINE ===

Date: 2008-07-22 (Eastern Daylight Time, UTC-04:00)

02:00:44 - Gmail session begins for jcoachj@gmail.com on device 192.168.15.4
Cookie: OL_SESSION=jcoachj@gmail.com-cal
Cookie: gmailchat=jcoachj@gmail.com/475090

02:01:26 - Device 192.168.15.4 visits sendanonymousemail.net (prior attempt)
Frame: 80614

02:02:57 - POST to www.sendanonymousemail.net/send.php (prior attempt)

02:03:43 - Device visits www.willselfdestruct.com
GET /secure/submit

02:04:24 - HARASSMENT POST to www.willselfdestruct.com/secure/submit
Frame: 83601
Client: 192.168.15.4:36044
Server: 69.25.94.22:80
MAC: 00:17:f2:e2:c0:ce
POST body: to=lilytuckrige@yahoo.com
subject=you can't find us
message=and you can't hide from us.
Stop teaching.
Start running.

02:04:24 - Server responds 302 Moved Temporarily
Location: http://www.willselfdestruct.com/secure/success
Set-Cookie: JSESSIONID=C16BFAC8710DA82AB3D8D80AE0F4F05A

02:04:55 - Gmail session continues on same device
EOF
cat reports/final_timeline.txt
=== NITROBA HARASSMENT CASE - FORENSIC TIMELINE ===

Date: 2008-07-22 (Eastern Daylight Time, UTC-04:00)

02:00:44 - Gmail session begins for jcoachj@gmail.com on device 192.168.15.4
Cookie: OL_SESSION=jcoachj@gmail.com-cal
Cookie: gmailchat=jcoachj@gmail.com/475090

```

Figure 16: Comprehensive Forensic Timeline Log — Compiling reports/final_timeline.txt logging suspect actions from 02:00:44 to 02:04:55 EDT.

```
Session Actions Edit View Help
- Prior attempt (sendanonymousemail.net): 02:02:57 EDT
- Harassment POST: 2008-07-22 02:04:24 EDT (frame 83601)
- Server response: 02:04:24 EDT (302 redirect)
- Gmail session continues: 02:04:55 EDT

Q8: What conclusion can you defend?
_____

CONCLUSION (Confidence: HIGH):
The device at 192.168.15.4 (MAC 00:17:f2:e2:c0:ce) submitted the harassment message
to www.willselfdestruct.com at 2008-07-22 02:04:24 EDT. The same device was
concurrently logged into Gmail as jcoachj@gmail.com, which corresponds to
Johnny Coach, a student on the Chemistry 109 roster.

LIMITATIONS:
1. Shared open wireless network: Multiple users share 192.168.15.4 (NAT).
2. The MAC address alone cannot prove identity; session cookies provide the link.
3. The Facebook cookie (beth@bethr.org) on the same network shows another user was active.
4. No direct forensic examination of the physical device was possible.
5. The User-Agent (IE 6.0 on Windows XP) does not match the Apple MAC prefix (00:17:f2) -
possible MAC spoofing or virtualization, though the MAC was consistent.

ALTERNATIVE EXPLANATIONS:
- Johnny Coach may have loaned the device to someone else.
- Another user may have accessed the Gmail session on the same device.
- The MAC could have been spoofed.
However, none of these alternatives are supported by the captured evidence; the
simplest explanation (Johnny Coach sent the message) is the most defensible.
EOF
cat reports/investigation_answers.txt
=== SBT-DF204 CASE STUDY 1 - INVESTIGATION ANSWERS ===

Q1: What file did you acquire, and how did you preserve its integrity?
_____

Acquired: nitroba.pcap from digitalcorpora.org (NITROBA University Harassment Scenario)
File size: 54 MB (94,410 packets)
SHA-256: 2b77a9eaeafc1d6af163d1ba793c96dbccacb04e6befdf1a0b01f8c67553ec2fb
Preservation: Created working copy (nitroba_working.pcap); hashes matched exactly.
Method: cp --preserve=timestamps; sha256sum verification

Q2: Which client system contacted the web service?
_____

Client IP: 192.168.15.4
Client Port: 36044
```

Figure 17: *Defensible Conclusion & Limitations Assessment* — Reviewing high-confidence attribution, alternative explanations, and open-network limitations in investigation_answers.txt.

```
Session Actions Edit View Help
possible MAC spoofing or virtualization, though the MAC was consistent.

ALTERNATIVE EXPLANATIONS:
- Johnny Coach may have loaned the device to someone else.
- Another user may have accessed the Gmail session on the same device.
- The MAC could have been spoofed.
However, none of these alternatives are supported by the captured evidence; the
simplest explanation (Johnny Coach sent the message) is the most defensible.

(kali@kali)~[~/SBT-DF204-CaseStudy1]
$ cd ~/SBT-DF204-CaseStudy1
cat > reports/evidence_log.txt << 'EOF'
=== EVIDENCE LOG ===

E01 | Frame 83601 | HTTP POST to willselfdestruct.com/secure/submit containing harassment message | Proves message content and origin | Appendix A
E02 | TCP Stream 1707 | Full TCP conversation between 192.168.15.4:36044 and 69.25.94.22:80 | Shows request/response, User-Agent, Referer | Appendix B
E03 | eth_src = 00:17:f2:e2:c0:ce | MAC address for device 192.168.15.4 | Links all activity from this IP to one device | Appendix C
E04 | Cookie: OL_SESSION=jcoachj@gmail.com-cal | Gmail session cookie from same device, 02:00:44 - 02:04:55 | Links device to identity jcoachj@gmail.com | Appendix D
E05 | Cookie: gmailchat=jcoachj@gmail.com/475090 | Gmail Chat session from same device | Corroborates Gmail session identity | Appendix D
E06 | Roster match: "Johnny Coach" | Chemistry 109 roster entry | Links jcoachj to a known student | Appendix E
E07 | SHA-256 of capture | 2b77a9eaeefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb | Evidence integrity | Appendix F
E08 | Cookie: beth@bethr.org (Facebook) | Facebook session from same network | Indicates another user on shared network (red herring) | Appendix G
EOF
cat reports/evidence_log.txt
=== EVIDENCE LOG ===

E01 | Frame 83601 | HTTP POST to willselfdestruct.com/secure/submit containing harassment message | Proves message content and origin | Appendix A
E02 | TCP Stream 1707 | Full TCP conversation between 192.168.15.4:36044 and 69.25.94.22:80 | Shows request/response, User-Agent, Referer | Appendix B
E03 | eth_src = 00:17:f2:e2:c0:ce | MAC address for device 192.168.15.4 | Links all activity from this IP to one device | Appendix C
E04 | Cookie: OL_SESSION=jcoachj@gmail.com-cal | Gmail session cookie from same device, 02:00:44 - 02:04:55 | Links device to identity jcoachj@gmail.com | Appendix D
E05 | Cookie: gmailchat=jcoachj@gmail.com/475090 | Gmail Chat session from same device | Corroborates Gmail session identity | Appendix D
E06 | Roster match: "Johnny Coach" | Chemistry 109 roster entry | Links jcoachj to a known student | Appendix E
E07 | SHA-256 of capture | 2b77a9eaeefc1d6af163d1ba793c96dbccac04e6befdf1a0b01f8c67553ec2fb | Evidence integrity | Appendix F
E08 | Cookie: beth@bethr.org (Facebook) | Facebook session from same network | Indicates another user on shared network (red herring) | Appendix G

(kali@kali)~[~/SBT-DF204-CaseStudy1]
```

Figure 18: Final Forensic Evidence Log Verification — Viewing complete evidence catalog (E01 to E08) confirming evidentiary chain of custody.

Student Declaration

I confirm that I completed this investigation in an authorized environment, preserved the evidence, did not target any third-party system, and accurately documented my own commands, observations, and conclusions.

Student Signature: Almustapha Yusuf

Date: 5/10/2026